Chapter 14

Indirect Price Discrimination

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Learning Objectives

- 14.1 Explain conditions in which sellers practice indirect price discrimination.
- 14.2 Discuss how sellers can prevent cannibalization of high-priced products using damaged good strategy and metering strategy.
- 14.3 Identify pricing strategies that allow firms to extract more consumer surplus by encouraging customers to self-select based on willingness to pay.
- 14.4 Explain how bundling different goods together can generate profits.

Chapter 14 – Main Points

- When a seller cannot identify low- and high-value consumers or cannot prevent arbitrage between two groups, it can still discriminate, but only indirectly, by designing products or services that appeal to groups with different price elasticities of demand, who identify themselves based on their purchasing behavior.
- Metering is a type of indirect discrimination that identifies high-value consumers by how intensely they use a product (e.g., by how many cartridges they buy). In this case, charge a big markup on the cartridges and a lower markup on the printer.
- If you offer a low-value product that is attractive to high-value consumers, you may cannibalize sales of your high-price product.
- When pricing for an individual customer, do not bargain over unit price. Instead, you should
    - Offer volume discounts;
    - Use two-part pricing; or
    - Offer a bundle containing a number of units.
- Bundling different goods together can allow a seller to extract more consumer surplus if willingness to pay for the bundle is more homogeneous than willingness to pay for the separate items in the bundle.

Warm-Up

- What product have you bought where the cheaper version felt frustratingly limited? Why do you think that version exists?

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Indirect Price Discrimination

- Indirect price discrimination occurs when a seller cannot directly identify high and low value consumers
- In these instances, the seller has to identify some feature of the product that is correlated with value.
  - Apps for phones and websites use these types of schemes. They offer a higher price to open up more of the app/website for those users who need/want the other features

- The key to indirect price discrimination is:
  - Make the cheaper version (the one the low value user wants) as unattractive as possible to the high value user.
  - If you don't do this, then the high value user will view the two products as not that much different, and enjoy more surplus out of the cheaper version

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Indirect vs. Direct Price Discrimination

Indirect price discrimination occurs when firms can’t directly identify customer types, so they design products or pricing options that lead customers to self-select based on their willingness to pay.

When to Use Each?

Can you observe customer type directly?	**Strategy to Use**
Yes (e.g., ID, location, usage)	Use direct price discrimination
No (e.g., unknown income or preferences)	Use indirect price discrimination

Indirect vs. Direct Price Discrimination

- Indirect price discrimination is best when:
  - You serve many diverse customers but can't observe who is who
  - You can design product versions or price menus that:
    - Attract high willingness to pay customers to premium options
    - Attract low willingness to pay customers to basic/lower-tier options

- Examples:
  - Airline seats (business v. economy)
  - Streaming plans (basic, standard, premium)
  - Software (home, pro, enterprise)
  - Fast food combos (meal bundles v. a la carte)

- Let customers reveal themselves

Discussion Activity 1

- If you can't tell which customers are high value, how would you design your pricing to let them reveal themselves?
- Think about “cheap but inconvenient” vs. “premium but easy” options. What types of design choices make the distinction clear?

Volume Discounting

- **Volume discounting** occurs when a firm charges less as more is bought.
- From a consumer standpoint, the consumer will gradually erode all their willingness to pay as they purchase more of a good.
  - A *two-part tariffs* (two-part pricing) is a way to make this happen.
  - For example, many colleges and professional football teams charge a flat fee (variable for the seat) and then charge for ticket prices on top of that. The charge for tickets is the marginal cost and the flat fee is the fixed cost.
- From a producer standpoint, purchasing more of a good (resource) is better as it lowers the average cost of production.
  - This is called an external economies of scale. It allows the firm lower average costs as output expands.

Tie-in Sales

- There are two types of tie-in sales:
  - **Requirement tie-in sales** – when customers who buy one product from a firm are required to make all their purchases of another product from that firm
  - **Package tie-in sale** – combining two goods and selling them together, so the customer cannot buy them separately (also known as bundling)
    - Under package tie-in sales the firm attempts to maximize profit by determining the reservation prices of their consumers. If reservation prices are known, then the firm can bundle the goods as opposed to selling them individually.

- Requirement tie-in sale can often be viewed in office copiers, Windows and Internet explorer
  - Can be used to identify heavy users of the product and charge them more
  - Goods and services purchases will differ across consumers, so the per-unit price of each differs as well.
  - The FTC challenged Sun Company of tying its blood-monitoring service to its schizophrenia drug, as the drug maker had a monopoly on the drug.
  - Perhaps, one of the most famous tying examples is when the FTC challenged Microsoft over tying Internet Explorer to Windows 98.

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Bundling

- Bundling is easily seen with printers, as printers come with ink cartridges. Other examples include: cable and satellite TV packages—who really wants QVC when they only want HBO?
    - **Bundling** is practiced by charging consumers different prices to maximize profits
        - Products are offered together for convenience or discount, but still optional.
    - **Pure bundling** is when firms don't allow the purchase of the items individually
        - Tennis shoes and shirts are common examples.
    - Firms know that consumers value the goods in the bundle differently, so one good has more value for one consumer than the other consumer and vice versa for the other good.
    - Bundling also requires that a firm can prevent resale.

- A last form of bundling is **mixed bundling**. In this scenario, the firm sells the items separately and together, but the bundle comes with a discount—think fast food value meals.
    - Microsoft Office is another mixed bundle of goods, as you can buy the programs separately.

Bundling

- When bundling works best:
  - Products are not perfect substitutes
  - Customer valuations are negatively correlated
  - Costs are low to combine or distribute together
  - Resale between customers is not possible
- Examples:
  - Netflix – movies and series
  - Microsoft Office – Word, Excel, PowerPoint, etc.
  - Food – combo meals
  - Travel – hotel and airfare package deals

- It works because it reduces variation in customer valuations and lets the firm capture more profit.

Bundling Example

	Consumer A	Consumer B
Good 1	$60	$65
Good 2	$15	$10

If sold separately, what price maximizes revenue for good 1? Good 2?

$60 $10

Assume zero costs, what is the total profit for the firm if the goods are sold separately?

Good 1 = $120 Good 2 = $20 profit = $140

What is the best profit maximizing strategy for the firm?

Bundle the goods for $75 for a total profit of $150

Discussion Activity 3

- Why do so many digital services (like Microsoft Office or streaming) offer bundles instead of selling each item separately?
- Consider customer value variation and the cost of individual sales. When is bundling more profitable?

Self-Selection Pricing

- When firms can't identify customer types directly, they design pricing structures that lead customers to reveal their willingness to pay by the choices they make.
  - This allows firms to extract more consumer surplus
- Summary

Strategy	How It Works	Real-World Example	Why It Extracts Surplus
Volume Discounts	Lower unit price for larger quantity	Costco, cell data plans	High-WTP customers buy more
Two-Part Tariffs	Upfront fee + per-use charge	Gym memberships, Amazon Prime	Locks in commitment; charges for use
Versioning / Bundling	Offer multiple product tiers or combos	Cable plans, fast food meals	Consumers self-sort by value

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Self-Selection Pricing

- Examples:
  - Gym memberships (two-part tariff)
    - Pay $50/month + $ for a class or trainer session
  - Cloud storage plans
    - 1-5 GB free then $1.99 for 50 GB and $4.99 for 100 GB
  - Fast food combos
    - 12-count nuggets, fries and drink at a discount
    - Customer pays more for each individually

- Self-selection strategies let customers tell you how much they’re willing to pay, and ensure you capture more value from each segment.

Damaged Goods v. Metering Strategies

- When firms offer multiple versions of a product, they must protect the premium version from being undercut by the cheaper one.
  Two key tools to manage this are
- Damaged goods – intentionally reduced quality or features on the lower-priced version so high-willingness-to-pay customers choose the premium versions
  - Slower performance (printers with software limits)
  - Lower durability (luggage with cheaper material)
  - Feature restrictions (Adobe free version vs Adobe pro)
  - Ads or interruptions (Mobile apps and games)
  - Goal: make low-price version unattractive to premium users at little added cost
- Metering strategies – charge based on usage so high-value customers pay more
  - Ink or toner (printers with cheap hardware and expensive cartridges)
  - Utility pricing (electricity and water)
  - Goal: let customers self-select into tiered usage

Activity – Build a Tiered Pricing Model

Scenario:

- You're working with a new software startup offering a cloud-based design tool. They want to design a 3-tier plan that extracts as much value as possible from users with different willingness to pay.

Instructions:

1. Propose three price tiers (basic, pro, premium).
2. List at least one feature restriction or friction for each lower tier that would make higher-value users self-select into paying more.
3. Describe how you'd use metering or two-part pricing to segment users by intensity.
4. Identify any risks (e.g., over-segmentation, backlash) that could hurt adoption or perception.

Activity – Build a Tiered Pricing Model

1. Propose tiers and
2. Feature restriction
   - Basic: $0 - $9, watermark, limited exports, ads, fewer templates
   - Pro: $19, no ads, more exports, access to collaboration tools
   - Premium: $49, full feature set, integrations, analytics, team set
3. Segmenting markets
   - Metering Strategy: Base plan free, but $/GB of storage or $/exported file
   - Two-Part Pricing: $10/month access fee + $2 per user or usage-based add-on
4. Risks
   - Too much restriction in lower tier
   - Cannibalization of higher tiers if lower tiers offer too much value
   - Lack of transparency in pricing

Discussion Activity 3

- Is it ethical for a company to intentionally make a product worse (e.g., slower or more ads) to steer people to a better version?
- What are the tradeoffs between accessibility and profit maximization?

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Summary: Indirect Price Discrimination Strategies

Strategy	How It Works	Goal	Examples	Risks/Tradeoffs
Metering	Tie price to usage or quantity consumed	Extract more surplus from high users	Ink cartridges, electricity pricing	Can feel punitive; complex tracking
Bundling	Sell products together (pure or mixed) to appeal to different WTP levels	Price based on buyer preferences across goods	Microsoft Office, cable TV packages	May confuse customers or reduce transparency
Damaged Goods	Intentionally degrade the low-end version to nudge high-WTP buyers upward	Self-selection through quality tradeoffs	Airline seat classes, slow printers	May appear manipulative or wasteful
Two-Part Tariff	Charge a fixed fee plus variable per-unit price	Capture both access and usage surplus	Costco membership, amusement parks	Can deter low-usage customers
Versioning	Offer product variations by quality/features	Let users self-select their WTP segment	Adobe Creative Cloud tiers, streaming plans	Hard to balance cost vs. perceived value
Tie-In Sales	Sell base product cheap, profit from required complements	Lock-in and meter use through complementary goods	Razor + blades, game consoles + games	Regulatory scrutiny; can provoke backlash

Case Study: Adobe Creative Cloud

- Adobe historically sold software as individual perpetual licenses (e.g., Photoshop for ~$700).
- In 2013, it launched Adobe Creative Cloud, shifting to a subscription-based model.
- Offers multiple pricing tiers:
    - Free limited tools (e.g., Adobe Acrobat Reader)
    - Single App plan (~$20/mo)
    - Full Suite plan (~$55/mo)
    - Student/educator discounts and business plans

Case Study: Adobe Creative Cloud

Strategy	How Adobe Uses It
Bundling	Creative Suite combines apps that vary in popularity (e.g., Photoshop + InDesign).
Self-Selection	Freelancers may choose individual apps; agencies go for full suite.
Damaged Goods	Free apps with fewer features and no cloud access encourage upgrades.
Metering	Cloud storage and export/rendering limits create “usage-based” price discrimination.
Tiered Pricing	Discounted student/education versions; business accounts pay more for admin controls.

Outcomes

- Increased Revenue Predictability: Subscriptions smooth out cash flow and reduce piracy.
- Higher Lifetime Value (LTV): Monthly payments lead to higher overall spending over time.
- Market Segmentation Success: Maintains high-end revenue while still serving price-sensitive users.
- Controversy: Some professionals disliked loss of ownership and inflexible pricing.

Is Adobe’s use of indirect price discrimination a fair way to segment its customer base?
Should they allow more customization (e.g., choose 3 apps at a lower price)?

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Antitrust Policy

- Overview of Antitrust Law
  - Market dominance is no offense.
  - Unfairly gained competitive advantage is illegal.
- Sherman and Clayton Acts
  - Sherman Act forbids restraints of trade and “monopolizing.”
  - Clayton Act focuses on mergers, interlocking directorates, price discrimination, and tying contracts.
- Antitrust Enforcement
  - Department of Justice pursues criminal violations.
  - FTC litigates civil violations.

Brief History of the Birth of Antitrust Policy

- Antitrust policies came about in the late 19th century as technological breakthroughs and economies of scale allowed firms to grow larger
- Firms began to form **trusts** (each firm transferred its voting shares of stock to a single board of trustees, and the trustees made a decision for the entire group of firms)
  - Examples include oil, kerosene, meatpacking, sugar, whiskey, lead, coal and tobacco
  - Organizers of trusts were called robber barons
  - Trust agreements were a matter of public record
- A change in a New Jersey statute in 1889 allowed firms to obtain a corporate charter in New Jersey and become “holding companies”
  - This allowed collusion

Sherman Antitrust Act (1890)

- Section 1: made any contract or combination in restraint of trade (e.g. price fixing) illegal

- Section 2: made any attempt to monopolize a market illegal.
  - AB Inbev managed to acquire SABMiller even though market share for both in the US was about 58%. The reason the deal was okayed is likely to do to divestiture of assets in the US.

- Later Amendment: Penalties for misdemeanors include $50,000-$1,000,000 fines for corporations and $50,000 fine and 1 year in prison or a $100,000 fine and 3 years in prison for individuals.

- Injured parties have the potential to get triple the damages in civil suits.

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Sherman Antitrust Act Early Cases

- The first test of the Act was against E.C. Knight and Company in 1895.
- The American Sugar Refining Company had acquired the stock of E.C. Knight and Company and three other sugar refining companies, bringing its total market share from 65% to 98%
- The decision of the Supreme Court ruled in favor of the sugar trust as the company was engaged manufacture and not interstate commerce
  - This decision gave the impression that mergers were legal and an effective way to eliminate price competition

- In 1898, the ruling of *United States v. Addyston Pipe and Steel Company* stated that collusive agreements between firms competing with one another was illegal, but still allowed mergers to continue to be legal
  - This ruling sparked the Great Merger Wave (1898 – 1902)
  - The irony is that this concentration movement was spurred on by seeking monopoly power more so than the first one

Standard Oil and American Tobacco

- Theodore Roosevelt's 1904 presidential campaign promise was to bust trusts
  - *Northern Securities Co. v. United States* was, perhaps, his greatest victory.
- Two prominent companies to be found in violation of the Sherman Antitrust Act were Standard Oil and American Tobacco
- The court decision was based on the "rule of reason"
  - Here the possession of monopoly power isn't illegal, but the acquisition of power through restraining trade is. The monopoly could also be legal if it had acquired its status via legal means
- Standard Oil and American Tobacco were found guilty based on their actions and not market size
  - Standard Oil was deemed to have unfairly obtained rebates from railroads and engaged in predatory pricing, *i.e*. they set prices below their average variable costs to drive out competition
  - American Tobacco was also deemed to have engaged in predatory pricing
  - Their sentence was to dissolve into smaller companies

Clayton Act (1914)

- Section 2 makes it illegal for sellers to price discriminate to buyers when doing so lessens competition or creates a monopoly. Otherwise, it is permissible if it is based on differences in grade, quality or quantity sold.
- Section 3 makes it illegal for sellers to lease, sell, or contract for the sale of a commodity on the condition that the lessee or buyer does not purchase, lease, or deal in the commodity of a competitor, if such an exclusive or tying contract substantially lessens competition or tends to create a monopoly.
  - The 1949 *Standard Stations* opinion helped to solidify this point
- Section 7 makes it illegal for a corporation engaged in commerce to acquire the stocks of a competing corporation or the stocks of two or more corporations that compete with one another if such intercorporate stock holdings substantially lessen competition or tend to create a monopoly.
- Section 8 makes it illegal for the same individual to be on the board of directors of two or more corporations (interlocking directorate) if the corporations are competitive and if any has capital, surplus, or undivided profits in excess of $1 million.

Other Antitrust Acts

- **Federal Trade Commission Act** (1914)
  - "unfair methods of competition were unlawful"
    - Focused on firms
  - Established the FTC which prosecutes violators
- **Robinson Patman Act** (1936)
  - Made illegal to sell more cheaply to one buyer or market than to others with intent to destroy/erode competition (price discrimination)
    - Price discrimination among firms – basically it protects smaller retailers against larger retailers
- The *per se* decision criteria became the way cases were settled.
  - *per se* – in itself
  - It would be used as a way to claim firm size in an industry is sufficient enough to rule against it in an antitrust suit

Other Antitrust Acts

- **Wheeler-Lea Act** (1938)
  - Forbade false or deceptive ads on food, drugs, etc.
    - Amended Section 5 of the FTC Act to protect consumers
- **Celler-Kefauver Antimerger Act** (1950)
  - Illegal to acquire the stock and assets of a competing corporation if it substantially lessened competition and tended towards a monopoly.

Alcoa (Aluminum Company of America)

- Late in the 19th century, Alcoa acquired a patent for a method to remove oxygen from bauxite set to expire in 1906.
- By 1903, Alcoa came up with a new method and new patent that would expire in 1909.
- Alcoa entered into contracts that prohibited bauxite producers from selling bauxite to any other American firm.
- In 1912, the courts deemed this illegal.
- To circumvent this, Alcoa always expanded production and manipulated prices (limit pricing)
- Alcoa maintained a monopoly type status until WWII when Reynolds and Kaiser (R&K) entered the market.
- Even after, Alcoa acquired R&K, prices have fallen largely due to the demand and supply of aluminum from the rest of the world.
- Source

Key Takeaways

- Indirect price discrimination is used when firms can't directly identify buyer types.
- Customers reveal their willingness to pay by selecting from designed product tiers.
- Strategies include:
    - Damaged goods: limit the low-end version to protect high-end revenue (e.g., free apps with ads)
    - Metering: charge more as usage increases (e.g., ink cartridges, cloud storage)
    - Self-selection: let customers choose based on value, not identity
- Bundling improves profits when:
    - Products are not perfect substitutes
    - Customer values for items are negatively correlated
    - Resale is not feasible
    - Distribution cost is low
- Tie-in sales and volume pricing also segment customers by intensity of use or purchase
- These strategies help extract more consumer surplus by customizing options to each segment

Knowledge Check

1. What makes indirect price discrimination different from direct?
2. What is a damaged goods strategy?
3. Give an example of metering and explain how it works.
4. What are two types of tie-in sales?
5. Why does bundling help firms increase profit?

Knowledge Check Answers

1. The seller cannot observe buyer types directly, must let customers self-select.
2. Offering an intentionally limited version of a product to preserve demand for a premium version.
3. Inkjet printers: hardware is cheap, but ink is expensive, heavy users pay more over time.
4. Requirement tie-in and package tie-in sales.
5. Bundling reduces variation in WTP and allows sellers to capture more consumer surplus.

Consultant’s Challenge: Let Customers Reveal Themselves

Scenario:

- A tech firm wants to grow revenue without alienating customers. They can’t directly observe buyer types but suspect wide variation in willingness to pay. You’re brought in to help them design an indirect price discrimination strategy.

Your Task:

1. How would you structure pricing tiers or bundles so that high-value users naturally choose higher-priced options?
2. What damaged goods, metering, or bundling tools would you recommend and why?